

Identifying U.S. Counties at Risk of Economic Decline

Using Public Socioeconomic Indicators
Capstone Project

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Economic Decline Is a Data Problem



What Economic Decline Looks Like

- Job loss and shrinking employment
- Declining household income
- Rising poverty rates

Why This Project Matters

- Helps governments design better, evidence-based policies
- Supports targeted investment decisions at the county level
- Enables data-driven planning for vulnerable communities

What We Set Out to Answer

Project Objectives

- Identify counties at high economic risk
- Understand how key indicators affect risk
- Build an interpretable risk score

Research Questions

Which counties are at highest risk?

How do Socioeconomic indicators influence risk?

Are there geographic patterns across regions?

Can these indicators be combined into a clear risk score?

U.S. Census Bureau — ACS 5-Year Estimates (2019–2023)

3,222

Counties

Represented in the dataset

~10MB

Data Size

Merged from multiple ACS tables

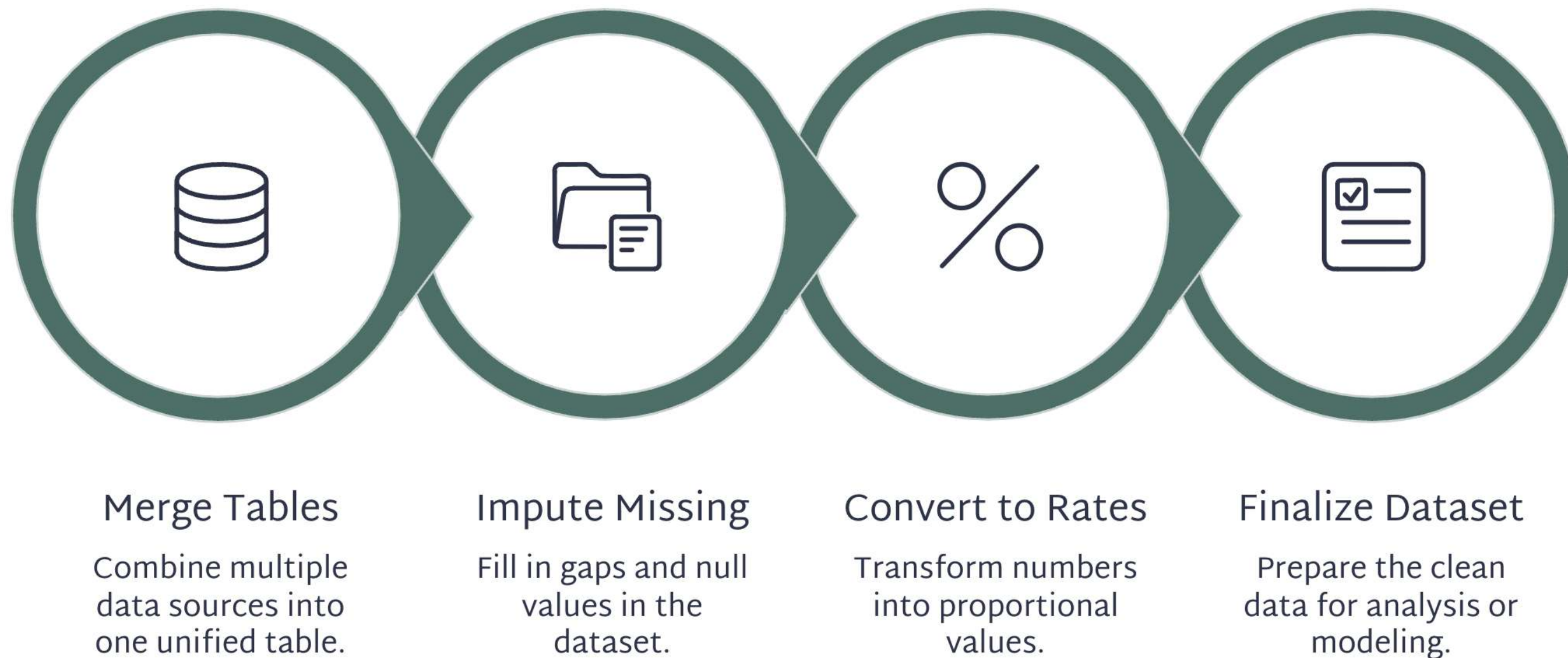
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Key Indicators

Income, poverty, unemployment,
education, homeownership

Multiple ACS tables were merged using `county_fips` into a single unified dataset. Each row represents one U.S. county, including a computed risk score and risk category.

DATA PREPROCESSING

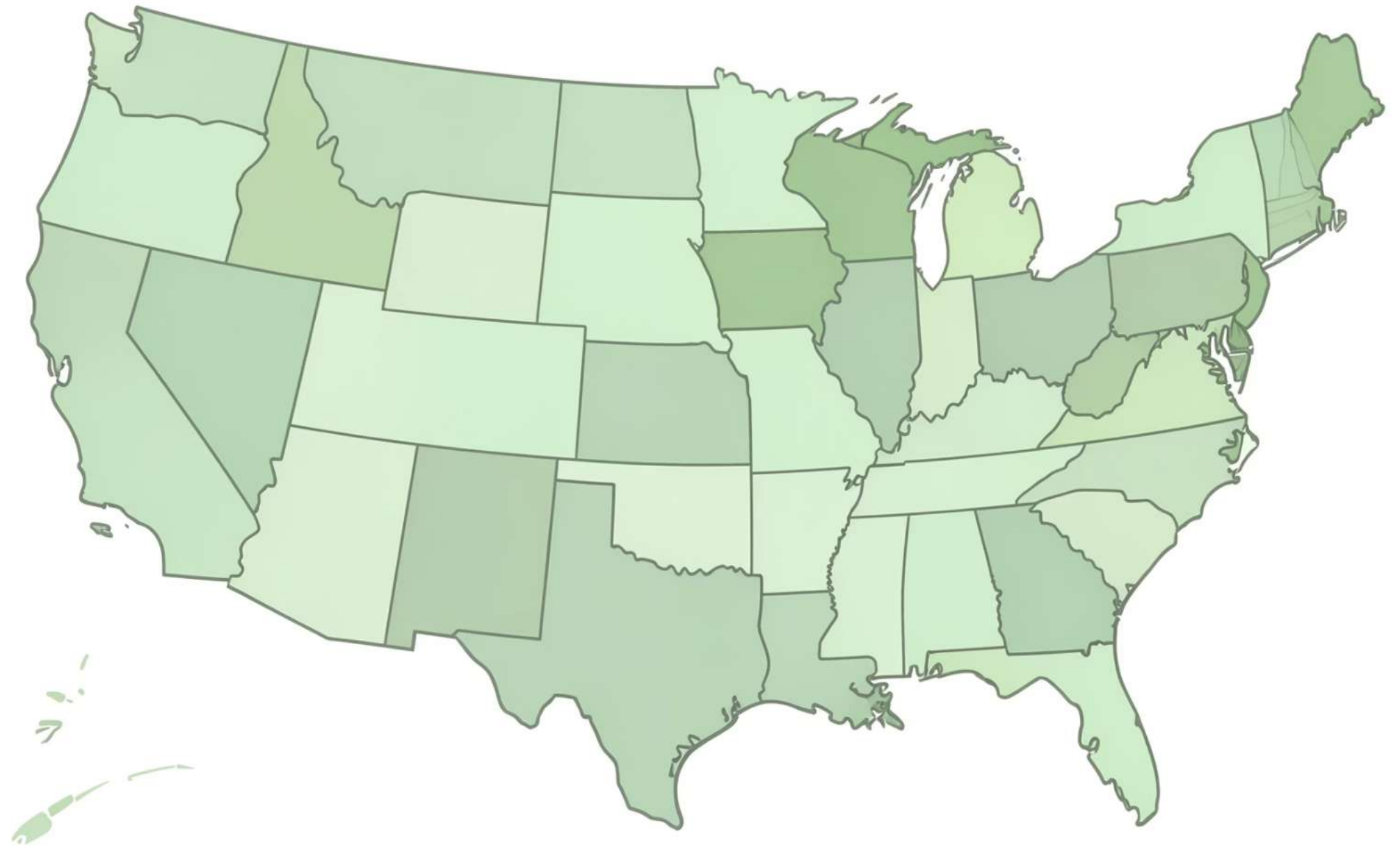


All indicators were standardized into rates to ensure consistent comparison across counties of varying population sizes. The official Census source resulted in minimal missing data.

Strong Variation Across U.S. Counties

Key EDA Observations

These patterns validate the selected indicators as meaningful signals of economic vulnerability.



Income varies widely across regions

Poverty and unemployment are strongly correlated

Education levels are closely linked to income

Clear **regional disparities** exist across the country

Building the Economic Risk Score

The Economic Risk Score is the **average of five normalised indicators**, each scaled to contribute equally. Income, education, and homeownership are inverted so that **all indicators represent higher risk** with higher values.



Income

Inverted - lower income = higher risk



Poverty

Higher rate = higher risk



Unemployment

Higher rate = higher risk



Education

Inverted - lower attainment = higher risk



Homeownership

Inverted - lower rate = higher risk

Counties are classified as **Low Risk**, **Medium Risk**, or **High Risk** based on their composite score.

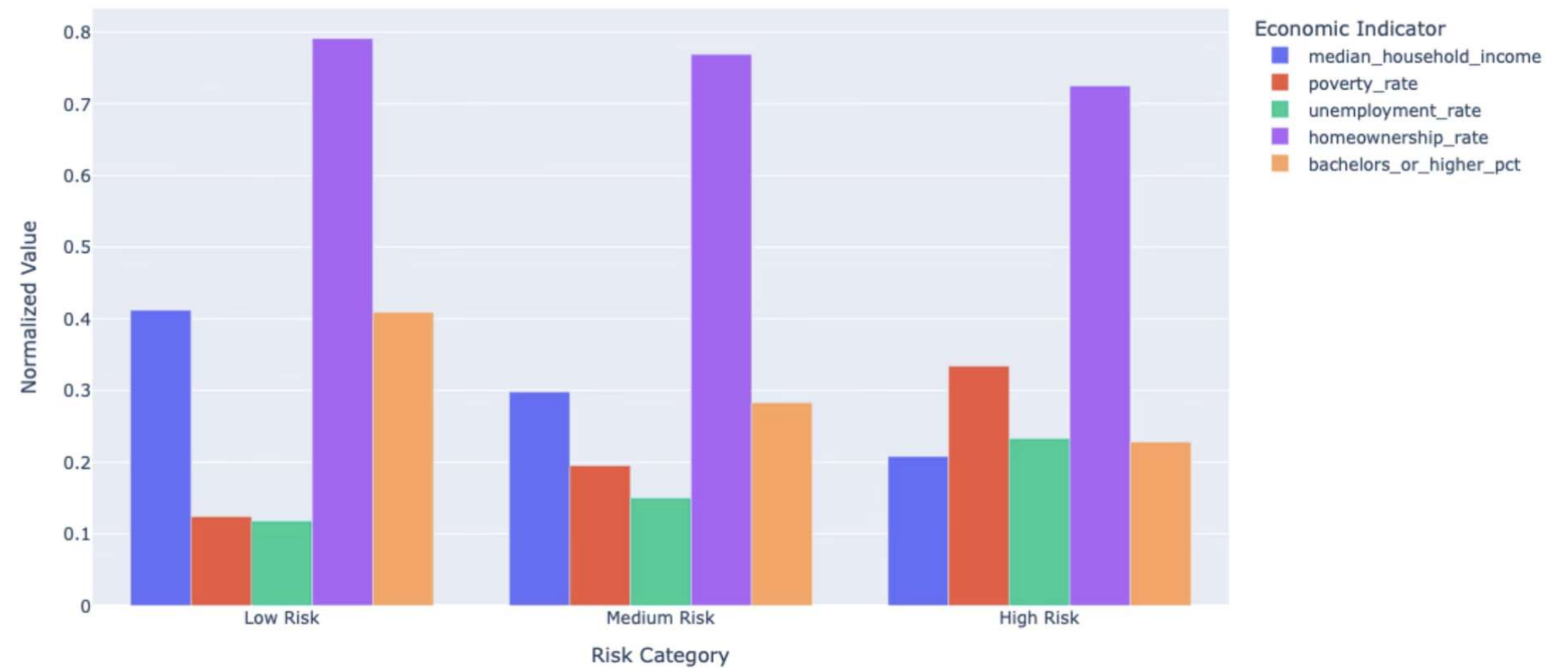
Clear Separation Across Risk Categories

High-risk counties consistently show:

- Lower median income and educational attainment
- Higher poverty and unemployment rates
- Lower homeownership rates

The clear separation across risk categories validates the composite risk score as a meaningful and interpretable measure of economic vulnerability.

Average Economic Indicators by Risk Category (Normalized)



Interactive County Risk Explorer



Dashboard Features



County-Level Map

U.S. map visualization of economic risk by county



Filtering Controls

Filter by state and risk category for targeted exploration



Summary Metrics

Side-by-side comparisons across risk groups and regions

What We Learned

- Economic conditions vary significantly across U.S. counties, highlighting strong regional disparities
- Counties with lower income and education levels consistently show higher poverty and unemployment rates
- Economic risk is influenced by multiple interconnected factors rather than any single indicator
- The composite risk score effectively combines these indicators into a clear and comparable measure
- This approach helps identify high-risk counties and provides useful insights for decision-making

FUTURE WORK

What's Next



Add more socioeconomic indicators (e.g., healthcare access, housing costs) for deeper insights



Analyse trends over time using historical data to track changes in economic risk



Apply machine learning models to predict future risk and identify emerging high-risk areas

